Huawei Access Node WFD-ACC-18 Hardware/Environmental Fault Resolution SOP at Whitefield POP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_5597

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for a Huawei Access Node (WFD-ACC-18) experiencing a hardware/environmental fault at the Whitefield POP, characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. The failure occurs due to the node being an access node with a Huawei vendor-specific implementation, which utilizes a custom ASIC design to handle forwarding and switching tasks. The ASIC's memory parity error detection mechanism triggers an alarm (ALARM_5597) of severity warning. The line card crashes are likely due to power supply redundancy loss, which compromises the node's ability to forward traffic. To mitigate this failure, it is essential to identify the root cause of the issue and take corrective actions to restore the node's functionality.

Pre-requisite Checks

- Pre-requisite: Verify redundant link status on the node

- Pre-requisite: Dispatch remote hands to the Whitefield POP to physically inspect the node

- Pre-requisite: Check the node's environmental parameters, including temperature, humidity, and power supply status

Potential Root Causes

Cause	Probability
ASIC memory parity error due to hardware failure	High
Line card crash due to power supply redundancy loss	High
Environmental factors, such as high temperature or humidity, affecting node performance	Medium

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include ASIC, Expected Output: ASIC configuration, Description: This command displays the current ASIC configuration, which can help identify any issues with the ASIC's memory parity error detection mechanism.

Command: display linecard | include crash, Expected Output: Linecard crash information, Description: This command displays information about the line card crashes, which can help identify the root cause of the issue.
```

```
Command: display power supply | include redundancy, Expected Output: Power supply redundancy status, Description: This command displays the power supply redundancy status, which can help identify if the power supply redundancy loss is the root cause of the issue.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace ASIC, Description: Replace the faulty ASIC with a new one to resolve the memory parity error issue.

Step 2: Step: Replace the crashed line card, Command: replace linecard, Description: Replace the crashed line card with a new one to resolve the line card crash issue.

Step 3: Step: Check and replace the power supply unit if necessary, Command: check power supply unit, Description: Check the power supply unit and replace it if necessary to resolve the power supply redundancy loss issue.

Post-Check Verification

```
Command: display current-configuration | include ASIC, Expected Output: ASIC configuration, Description: Verify that the ASIC configuration is correct and there are no issues with the memory parity error detection mechanism.

Command: display linecard | include crash, Expected Output: Linecard crash information, Description: Verify that the line card crash issue has been resolved.

Command: display power supply | include redundancy, Expected Output: Power supply redundancy status, Description: Verify that the power supply redundancy status has been restored.
```

Roll-back Procedures

- Step: Restore the original ASIC configuration, Command: restore ASIC configuration, Description: Restore the original ASIC configuration in case the replacement ASIC does not resolve the issue.

- Step: Restore the original line card configuration, Command: restore linecard configuration, Description: Restore the original line card configuration in case the replacement line card does not resolve the issue.

- Step: Restore the original power supply unit configuration, Command: restore power supply unit configuration, Description: Restore the original power supply unit configuration in case the replacement power supply unit does not resolve the issue.